higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING PRACTICE AND MAINTENANCE
NQF LEVEL 3

7 MARCH 2019

This marking guideline consists of 7 pages.

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QUESTION 1: TECHNOLOGICAL SOLUTIONS IN ENGINEERING

1.1 1.1.1 Manuals include step-by-step practical instructions on how to solve a particular engineering problem.

1.1.2 Text books usually contain theoretical or academic information about specific engineering problems and their solutions.

1.1.3 Technical reports are written by engineers or scientists explaining how they have solved a particular problem and whether or not their solution worked.

1.1.4 Manufacturer's specifications are clear statements, information and instructions from the manufactures of a product such as a machine, about the product features, what it is designed for, and how to use it.

(4 × 1) (4)

1.2
- Performance problems
- Safety problems
- Reliability problems
- Operating problems
- Maintenance problems (Any 4 × 1) (4)

1.3
- Modelling. Models help engineers to develop ideas into approximations or representations of real products.
- Computer simulations. CAD software allows us to develop imitations of the product.
- Prototypes. Prototypes are working models of the design solution.
- Pilot studies. Pilot your project to potential users. (4)

1.4
- The manufacturer will earn more money in the long run because it will be a better product.
- The improved or modified design will better satisfy customer needs and wants.
- It will be more reliable and will last longer.
- It will be easier for the customer to maintain and to repair.
- It will be cheaper to produce because the modification will use newer, cheaper parts apart form the latest materials and technology.
- It will be of a higher quality.
- It will be safer to people and to the environment.
- It will be easier to upgrade in future.
- It will compete more successfully against other products.
- It will be easier to package and distribute to customers.
- Overall, it will be a more successful product, process or system.

(Any 4 × 1) (4)

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1.5 1.5.1 When designing for quality, engineers focus on the products of the highest quality.

1.5.2 When designing for export, engineers focus on producing designs for the export market.

(2 × 1) (2)

1.6
- Modifications: Making either a few small adjustments to the design or adapting the entire product or solution.
- Upgrades: What is new today will be out of date tomorrow. Technology changes rapidly and, therefore, solutions must keep up with trends.

(2)
[20]

QUESTION 2: SYSTEMATIC FAULT

2.1
2.1.1 E
2.1.2 D
2.1.3 F
2.1.4 B
2.1.5 H
2.1.6 C
2.1.7 A
2.1.8 G

(8 × 1) (8)

2.2
- Question the customer
- Direct observation
- Trial-and-error replacement of parts

(3)

2.3
- Question the customer
- Direct observation
- Taking measurements
- Trial-and-error replacement of parts
- Manufacturer's servicing data and drawings
- Simulations
- Set up system to fail

(Any 5 × 1) (5)

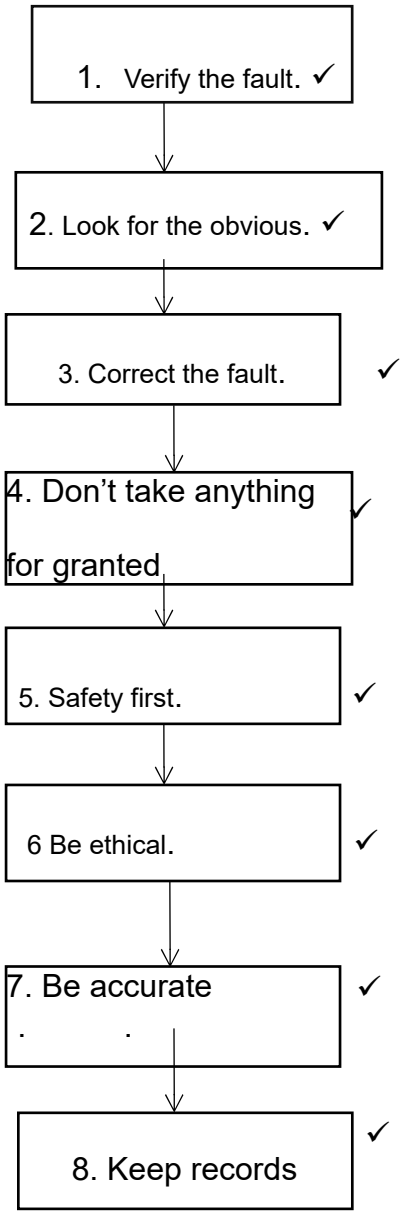
2.4 2.4.1
- Failure happens when a product or system does not perform as expected.
- A fault is when a part in the product or system has broken down.

2.4.2
- Relevant symptoms are those that give you a clue as to the symptoms of a fault.
- Irrelevant symptoms lead you away from the clues to the underlying fault.

(2 × 2) (4)
[20]

QUESTION 3: DIAGNOSIS OF MECHANICAL AND ELECTRICAL FAULTS IN ENGINEERING

3.1



(8 × 1) (8)

3.2
- Manual aids
- Technical aids
- Information aids

(3)

3.2 Step 1: Prepare and plan to repair the fault.
Step 2: Take all the necessary safety precautions including wearing the necessary protective clothing.
Step 3: Gather and prepare all the tools you will need.
Step 4: Before you take the equipment or the engine apart, make sure you have detailed diagrams of how it is assembled.
Step 5: Clean the area and the engine before you start, as specified by the manufacturer. (5 × 1) (5)

3.4
- Your workplace address
- Type of equipment
- Age of the equipment
- Tests completed
- Faults
- Recommendation action
- Your name and signature
- Name of the signature of supervisor (Any 4 × 1) (4)

[20]

QUESTION 4: LUBRICANTS AND LUBRICATION SYSTEM

4.1
- The hydrostatic principle says that there must be a balance of force against the two surfaces that are in contact with each other✓ and this is achieved by lubricating the two surfaces.✓
- The hydrodynamic principle says that lubricants are used to separate moving (dynamic) parts in a system✓ by creating a thin film or oil so that they do not wear through friction.✓ (4)

4.2 4.2.1 False
4.2.2 False
4.2.3 True
4.2.4 False
4.2.5 True
4.2.6 True
4.2.7 True
4.2.8 True
(8 × 1) (8)

4.3 4.3.1 High-pressure lubrication system. Pressure is applied to pump a lubricant into the system.✓ Pressure is a force that is exerted on a surface. Pressure is high or large when the weight of the force is concentrated on a small area.✓

4.3.2 Low-pressure lubrication system. Oil is fed under low pressure to the various parts of a machine, for example, a car's air-conditioning system.✓ It works on the basic principle that as a liquid changes into a gas, heat is absorbed.✓

4.3.3 Centrally distributed lubrication system. Some machines have a number of lubricating points to which lubricant must be delivered.✓ An oil pump operates under pressure to pump the lubricant from the reservoir through a network of pipes to these points.✓

(3 × 2) (6)

4.4
- A drop in the system pressure decreases the efficiency of the system.
- Leaks can force the equipment to cycle more frequently, shortening the lifespan of almost all system equipment.
- Increased maintenance reduces the time during which users have access to the system. (Any 2 × 1) (2)

[20]

QUESTION 5: MAINTENANCE OF PLANT AND EQUIPMENT IN ENGINEERING

5.1
- Suitability. Both the plant and its equipment must be suitable for the job.
- Efficiency. Plant and equipment should operate with efficiency.
- Reliability. Plant and equipment must be able to work reliably and in a satisfactory way.
- Safety. Equipment needs to be safe and operate in a safe way.
- Cost-effectiveness. The plant and equipment must operate in a cost-effective and economical way. (Any 4 × 1) (4)

5.2
- Housekeeping. Storing material and equipment properly.
- Fire protection. Have enough fire extinguishers equipment available in good working order.
- Planning the task. The task should be planned properly before starting.
- Medical and first-aid. Having health and safety representation.
- Training workers. Everyone should be trained to take care of hand tools and equipment and to use of PPE. (Any 4 × 1) (4)

5.3 STEP 1: Identify potential hazards so that you can stop accidents before they happen by doing something about these dangers.
STEP 2: Remove potential hazards.
STEP 3: Reduce potential hazards.
STEP 4: Report potential hazards. (4 × 1) (4)

5.4
- Most engineering equipment is potentially dangerous if you do not use it correctly.
- The equipment is likely to fail if it is not used for the correct purpose, or your job is likely to be unsuccessful.
- Prepare and set up the equipment properly before you begin your task and check that the equipment is in good working order.
- Use the equipment according to the manufacturer's specifications.
- It is very important to practice using the equipment until you can use it with confidence. Always remain alert and aware.
- Being able to identify potential hazards in equipment enables you take action to remove or reduce these hazards to ensure safety.
- Using the equipment also means cleaning it properly and maintaining it so that it is in good working order. (Any 4 × 1) (4)

5.5
- An engineering plant is a factory or building in which an industrial process happens✓ such as the manufacturing of a vehicle or an air-conditioning system.✓
- Engineering equipment includes all the tools and equipment used in the plant✓ to carry out the functions of the plant.✓ (4)

[20]

TOTAL: 100

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